

SYNOPSIS

On the evening of 7 June 2006, the general cargo ship *Skagern* and the container ship *Samskip Courier* collided in the Humber estuary in dense fog. Fortunately there were no injuries on either vessel. *Samskip Courier* sustained minor damage to her bow but was able to continue on her voyage to Rotterdam. *Skagern* was extensively damaged forward and required major repairs.

Both vessels had experienced pilots on board at the time of the accident, and *Skagern* in particular was a regular visitor to Hull.

Skagern arrived at Spurn light float, where she embarked her pilot at 2100. A master/pilot exchange of information was carried out and the vessel proceeded inbound towards King George Dock, Hull, at speeds of up to 11.5kts. Visibility was moderate, but it soon deteriorated, and at times dropped to less than a cable.

At 21:30, *Samskip Courier* embarked a pilot at King George Dock, and after leaving the dock proceeded seaward at speeds of up to 12.5kts, in thick fog.

The two pilots were aware that the vessels would meet each other at some point; they had talked to each other on mobile telephones, and VTS also informed them of each other's location. The vessels acquired each other on radar when some 2 miles apart, as they closed at a combined speed of over 23kts in visibility of as little as 1 cable. Neither vessel plotted the other on radar as they converged.

VHF radio communications between the two pilots, together with the radar images, revealed that the vessels were on a collision course. The subsequent attempts at emergency avoidance were unsuccessful, and the ships collided, head-on.

Samskip Courier sustained relatively little damage and was able to proceed to anchorage at the mouth of the Humber. However, *Skagern's* bow and collision bulkhead were both penetrated, and she started to sink as she attempted to reach King George Dock. To prevent the vessel from sinking in the channel, the master and pilot manoeuvred *Skagern* onto a mid-stream sandbank with the help of a tug. After further damage assessment, another tug was attached and *Skagern* was towed astern into King George Dock, where her bow then sank and settled on the basin bed before she could be emptied of her cargo.

MAIB inspectors visited both vessels and, as part of their investigation, were able to retrieve data from *Samskip Courier's* Voyage Data Recorder (VDR).

The ensuing investigation identified contributing factors to the accident; these included:

- Failure to apply long established collision avoidance methods by the masters and pilots of both vessels.
- Pilot /master relationships: the masters' over reliance on the pilots.
- Poor interaction and communications among the bridge teams.
- Loss of situational awareness by *Samskip Courier's* pilot.
- The positioning of Sand End light float.
- Use of mobile telephones on the bridge.

SECTION 1 - FACTUAL INFORMATION

1.1 PARTICULARS OF *SAMSKIP COURIER* AND ACCIDENT (Figure 1)

Vessel details

Registered owner	:	MS Swipall Jan Kahrs GmbH.
Manager	:	J.Kahrs Bereederung GmbH & Co. KG.
Port of registry	:	St John's
Flag	:	Antigua and Barbuda
Type	:	Container
Built	:	Romania, 2006
Classification society	:	Germanischer Lloyd
Construction	:	Steel
Length overall	:	140.59m
Gross tonnage	:	7852
Engine power and type	:	8400kW, MAK 9M43
Service speed	:	18 knots
Other relevant info	:	Thrusters (bow, aft) 700kW / 500kW. Single CP propeller, left handed. Draught, 5.6m.

Accident details

Time and date	:	22:58 on 7 June 2006
Location of accident	:	53° 42'.5N, 000°14'.2W. Sand End light float, Humber estuary
Persons on board	:	13
Injuries/fatalities	:	None
Damage	:	Minor damage to stem and bulbous bow.

1.2 PARTICULARS OF *SKAGERN* (Figure 2)

Vessel details

Registered owner	:	Ahlmark Lines A.B
Manager	:	Barber Ship Management, Norway
Port of registry	:	Karlstad
Flag	:	Sweden
Type	:	Cargo
Built	:	Japan, 1983
Classification society	:	Lloyds
Construction	:	Steel
Length overall	:	106m
Gross tonnage	:	4451
Engine power and/or type	:	2826kW Mitsubishi
Service speed	:	12.5 knots
Other relevant info	:	Bow thruster 500kW. Single CP propeller, right handed. Draught, 6.3m. Hull strengthened to Ice Class 1.
Persons on board	:	11
Injuries/fatalities	:	None
Damage	:	Major damage to stem. Penetration of collision bulkhead.

1.3 BACKGROUND

Samskip Courier was a new purpose built container vessel chartered by Icelandic vessel operators, "Samskip," from her German owners, MS Swipall Jan Kahrs GmbH. She was managed by German ship managers J.Kahrs Bereederung GmbH & Co. KG and registered in Antigua and Barbuda. She had been in service only 6 weeks since her delivery from her builder's yard, and during that time had been plying between Rotterdam and Hull.

Skagern was a 23 year old general cargo vessel owned and operated by Swedish company, Ahlmark Lines. She was managed by Barber Ship Management of Norway and was registered in Sweden.

Skagern's regular route was between Sweden and Hull or Rochester; she had been a regular fortnightly visitor to the Humber since 1987.

1.4 NARRATIVE

(All times are UTC+1 and all courses are true)

Much of the information in this report is derived from *Samskip Courier's* Voyage Data Recorder (VDR) and from VHF radio and radar recordings of Humber Vessel Traffic Service (VTS).

1.4.1 Environmental conditions

- Dark.
- Wind: NE light airs.
- Sea state: smooth.
- Poor visibility with dense fog at times.
- Neap tides; low water slack.

1.4.2 Events leading to collision

A copy of the chart of the collision area is shown in **Figures 3 and 4**; this report makes frequent references to navigational marks on these figures.

During the evening of 7 June 2006, the general cargo ship *Skagern* was inward bound from Skelleftehamn, Sweden, to Hull, with a cargo of timber and copper.

The vessel arrived at Spurn light float at 21:00, where a pilot was embarked. Visibility was moderate, about 2 to 3 miles, but was forecast to deteriorate; by the time *Skagern* reached the Bull buoy, visibility was down to 2 cables (0.2 nautical mile).

In addition to the pilot, *Skagern's* bridge team consisted of the master, second officer and an AB. The pilot had conduct of the ship and was steering by autopilot from the starboard side of the bridge console, with the master at the engine controls on the port side. The second officer was plotting positions, while the AB was acting as lookout.

As *Skagern* passed Grimsby, the lights of the town could be seen before visibility reduced to 1-3 cables again. There were no vessels visible on the radar; however, the pilot had informed the master that an outbound ship, *Samskip Courier*, was further up the estuary.

Skagern proceeded up the Humber estuary at 80% of ahead power, which gave a speed over the ground (SOG) of 10 - 10.5kts; despite the reduced visibility, no sound signal was sounded. As *Skagern* approached Immingham Oil terminal (IOT), the pilot requested a speed reduction to slow ahead (6 - 7kts) to comply with local regulations. Visibility at this time was 2 cables. Once past IOT, the pilot advised that speed could be increased again. Speed was duly increased to 80% ahead and, as the ebb tide slackened, her speed increased to 11 - 11.5kts. *Skagern* continued upstream, passing close to starboard hand buoys.

At 22:41 *Samskip Courier* left the lock at King George Dock, Hull. On the bridge were her master, a pilot, and an ordinary seaman (OS) acting as lookout and standby helmsman. The master conned the ship from the bridge wing while the pilot gave him advice.

Samskip Courier's pilot informed Humber VTS on VHF radio that they were clear of the lock, and also reported that the local visibility was between 1 and 2 cables. VTS replied that: they were clear to sail; the depth on the dock tide gauge was 1.7m; there was no traffic in the area and to expect similar reduced visibility all the way to the sea. When clearing the lock, the ship gave one prolonged blast on her whistle, as required by port byelaws. While *Samskip Courier* was being manoeuvred from the lock, her pilot received a telephone call from *Skagern's* pilot. However, as this was not a good time to speak, they agreed to talk again at a more convenient time.

Once *Samskip Courier* was in the river and heading downstream, her pilot asked for the steering to be put on autopilot, and took over the con of the ship from the port side of the bridge console. The master was at the starboard side of the console, while the OS was by the port wing acting as lookout/standby helmsman.

At the pilot's request, *Samskip Courier's* speed was increased to slow ahead (almost 9kts). Soon after this, the master asked if he could increase speed further, but was advised by the pilot not to do so until they passed Saltend jetties.

At 22:47, the pilot of *Samskip Courier* returned *Skagern's* pilot's mobile telephone call, and they spent approximately 1½ minutes talking about general issues and also discussing the vessels' respective positions and the visibility. At this time, *Skagern* was near buoy No15A, 4.5 miles downriver from *Samskip Courier*.

At 22:52 *Samskip Courier* passed the Saltend jetties, at 0.25 mile distance. Her pilot informed Humber VTS on VHF radio of their position and that the visibility was "fully 2 cables". Elbow buoy was also clearly visible. Soon after, when *Skagern* was at a range of 2.25 miles and its target was just painting onto *Samskip Courier's* radar screen, the pilot advised the master that the engine speed could be increased to half ahead. The ship's speed quickly built up to almost 13kts. *Samskip Courier* then passed close to the Elbow buoy on a course of 136°. Over the next 0.5 mile, the pilot gave a series of incremental alterations on the autopilot to gradually bring the ship's head around to 152° - steering on Paull Sand buoy. This gradual alteration allowed *Samskip Courier* to edge across to the inbound (north side) side of the channel. The pilot believed that the flood tide had set the ship onto the Elbow buoy, and therefore easing across to the north side of the channel would prevent *Samskip Courier* from doing the same thing as she passed the Sand End light float.

The pilot of *Samskip Courier* had the port radar on 1.5 miles range, north up, offset centre, relative motion display. The master's starboard radar was on similar settings and was being scanned between 0.75 mile and 1.5 miles range. The target of *Skagern* appeared on the radar screen at a range of 2.25 miles when she was approximately 6 minutes away. However, she was not positively plotted, and the pilot was unsure of her range when he first noticed her in the vicinity of the Paull Sand Buoy. Both radars had ARPA capability, but this was not activated on either set.

On *Skagern's* bridge, the port radar (master's) was set on predominantly 1.5 miles range, offset centre, north up, relative motion display; the starboard radar (pilot's) was on 0.75 mile, offset, north up display. Both radars were occasionally scanning up to 3 mile range. The radars had ARPA capability, but this was not employed.

When the vessel was close to buoy No17 (N Holme), *Skagern* picked up *Samskip Courier* on her port side radar at a range of 2.25 miles. At 22:55, *Skagern*'s pilot made a routine position call on VHF radio to VTS Humber, as they passed Paull Sand buoy. He also reported 1 cable visibility. VTS in turn informed *Skagern* that *Samskip Courier* was outward bound, and also gave the tide gauge reading at King George Dock. At this point, the ships were 1.2 miles apart and closing at a combined speed of over 23kts – 1 mile every 2½ minutes. The combined length of the two vessels was about 1.25 cables – a slightly greater distance than the prevailing visibility.

Fifteen seconds after hearing *Skagern*'s communication with VTS, *Samskip Courier*'s pilot requested a speed reduction from half ahead (12+ knots or 50% on the power delivery setting (PDS)) to slow ahead (almost 9kts; 25% PDS). However, instead of reducing to slow ahead, as requested, the master reduced to only 36% PDS, which gave a speed reduction of about 1kt. By this time, the ships were clearly visible on each other's radar. When 0.8 mile apart, the master of *Samskip Courier* put his radar's electronic bearing line (EBL) onto 162°, their next course heading after clearing Sand End light float, to establish if the ship was clear to alter to starboard onto the new heading (**Figure 5**). The EBL showed that they were, indeed, clear of the light float, and the master was expecting the pilot to bring her round at any second. When the ships were 0.62 mile apart, the master reduced speed further to 30% PDS, without telling the pilot. However, this was still in excess of the speed that the pilot had requested earlier.

Skagern's master monitored *Samskip Courier*'s approach on his radar and observed she was not altering to starboard to pass down their port side as he expected. However, with the pilot conducting the vessel, he was not unduly concerned.

At 22:57:34, when 4.25 cables apart and when the Sand End light float was abeam the ship's starboard bow - exactly 60 seconds before collision (**Figure 6**), *Samskip Courier*'s pilot called up *Skagern* on VHF channel 12 and enquired if they were keeping close to the red buoys. *Skagern*'s pilot informed him that he was on the north (appropriate) side of the channel. *Samskip Courier*'s pilot replied immediately, and confirmed that they were also on that same side of the channel. Eight seconds later, *Samskip Courier*'s pilot instructed her master to go to slow ahead and engage hand steering. After a further 3 seconds, *Skagern*'s pilot requested a change from VHF channel 12 to channel 10. *Samskip Courier*'s pilot became confused with the radio, and fumbled to find channel 10 and, as a result, voice contact was not made again before the vessels collided. In the remaining seconds before the collision, *Samskip Courier*'s pilot advised her master, who was now at the helm, to bring the vessel hard to port. The master queried this instruction by repeating it in a questioning fashion. He received confirmation from the pilot, and duly did as he was advised. Immediately after this, the pilot cancelled the instruction because he could now see *Skagern*'s red (port) light. Shortly afterwards, the ships collided.

In the final minute before the collision, *Skagern*'s master was extremely concerned by the radar image in front of him. Because his view forward was obstructed by the large deck cranes on his vessel's port side, he went out to the port bridge wing to try to get a visual sight of the approaching vessel (**Figure 2**). Peering through the fog he saw white lights, which turned out to be accommodation lights below *Samskip Courier*'s bridge. He rushed back into the wheelhouse, put the vessel into hand steering, and threw her hard to starboard at the same time as hitting the engine controls for full astern. The pilot shouted an instruction for the helm to be put hard to port as he could now see the other ship's starboard light. The master attempted this, but it was too late to be

1.6 EXTRACTS FROM VDR EVIDENCE

The following information is taken from *Samskip Courier*'s VDR and VTS Humber records.

From the time *Samskip Courier* cleared King George dock at 22:41, until the point of collision, a period of approximately 17 minutes elapsed. Other than the mobile phone conversation between *Samskip Courier* and *Skagern*'s pilots, at 22:47, all further verbal information about the vessels' positions etc. was conveyed through VTS until the radio communications between the two pilots in the final stages when the vessels were 8 cables apart, and approximately 60 seconds before collision. Other key events, and *Samskip Courier* radar images at the time, were as follows:

- At 22:52 *Samskip Courier* notified VTS that they were passing Saltend and the jetties were clearly visible at 2 cables; they then increased speed to half ahead (12.7kts). At this same time *Skagern* became visible on *Samskip Courier*'s radar at 2.25 miles range (**Figure 12**), approximately 6.5 minutes before collision.
- At 22:53:40 *Samskip Courier*'s pilot gave starboard helm, which was held for 50 seconds as the ship passed Elbow buoy (**Figure 13**). Closing speed was 23.4kts.
- At 22:54:44 *Samskip Courier*'s pilot counteracted the starboard swing by giving port helm which was held for 26 seconds (**Figure 14**).
- At 22:55:30 both ships were appraised on VHF channel 12 of their respective positions by VTS, approximately 3 minutes before collision (**Figure 15**). Closing speed was 23.5kts. Visibility less than 1 cable.
- At 22:55:50 *Samskip Courier*'s pilot requested a speed reduction to slow ahead; the master did not give the full reduction requested, or notify the pilot of this.
- At 22:56:39 the master of *Samskip Courier* applied the EBL to his radar to establish if they were clear to come round onto the next course (**Figure 16**).
- The first radio contact between the pilots took place at 21:57:34, 60 seconds before collision (**Figure 17**). *Skagern*'s pilot requested *Samskip Courier*'s pilot to change radio channels and requested a reduction in his own ship's speed. *Skagern*'s master went out onto the port bridge wing to try and get a visual sighting of *Samskip Courier*. Closing speed was 22.1kts.
- A period of about 20 seconds was spent by *Samskip Courier*'s pilot trying to select channel 10 on the radio (**Figure 18**). Closing speed 21.0kts.
- At 22:58:31 an evasive manoeuvre of hard starboard and full astern was attempted by *Skagern*'s master; an evasive manoeuvre to hard aport was initiated by *Samskip Courier*'s pilot.
- 22:58:34 collision (**Figure 19**).

1.17 TIDAL INFLUENCE

The tide on the evening of 7 June had a range of 4.2m. Low water was predicted at 22.18, with a predicted height of 1.9mm. Actual time of low water at King George Dock was 22:20, with a height of 1.7m. The mid channel tide can run for more than half an hour beyond the parallel shoreside tide. *Samskip Courier* cleared the lock at 22:41, 21 minutes after low water at the dock.

Low water depth at King George Dock tide gauge was chart datum plus 1.7m; therefore the depth on the river would have been charted soundings plus at least 1.7m.

After the collision, as *Samskip Courier* lay stationary in the water seaward of Sand End light float, the VDR showed her COG and SOG to be 265°, 0.3kts. This set and drift reflected the combined effect of the tide and wind at that time.

1.18 POSITION OF COLLISION

The collision took place 0.2 mile south-east of the Sand End light float (**ref. Figure 4**). This float marked what used to be the south eastern navigable extremity of Hull Middle, but over the years the seabed topography changed due to tides and currents. Sand End light float was positioned about 1.1 cables from the Hull Middle 7 metre contour line, which was recognised as a safe water guide in this particular area. From the light float to the 7 metre contour, on the north side of the channel, was some 2.0 cables. There was sufficient depth of water between the light float and the shallows it marked to allow many vessels to “slip” or pass the buoy on the wrong side, if need be. Sand End light float interrupted a straight line run from Elbow buoy to Skitter Haven buoy and, as such, created a slight ‘knuckle’ and constriction in the channel. It was therefore not the best place for two ships to meet. Many pilots recognised this, and often arranged or timed their meetings so that they did not happen at this point.

SECTION 2 - ANALYSIS

2.1 AIM

The purpose of the analysis is to determine the contributory causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

2.2 CAUSE OF THE ACCIDENT

The accident was primarily caused by a failure to apply long established collision avoidance procedures by the masters and pilots of both ships. *Samskip Courier* failed to keep to the starboard side of the channel as she approached and rounded Sand End light float. This error was exacerbated by the fact that both ships were travelling at an unsafe speed for the prevailing conditions, in circumstances which allowed little time, or room, for avoiding action when it became apparent that a serious situation had developed.

Poor bridge team interaction, exacerbated by poor communications between key persons, played a major role in the accident. This was highlighted by the masters' total trust in their pilots, and their reticence to take navigational control when it became apparent that an emergency situation was developing.

The pilot of *Samskip Courier* lost situational awareness at a critical moment which, in turn, contributed to ineffectual decision making culminating in the collision.

2.3 FATIGUE

Working hours for the bridge teams involved in this accident were not onerous, with adequate rest periods given between duty periods. Fatigue is not considered an issue for them. However, the pilot on *Samskip Courier* was suffering from a head cold and cough, which were preventing him from sleeping properly. There was, however, ample opportunity to prepare for late evening work by taking a nap beforehand, but he had difficulty in sleeping during the daytime. Nevertheless, by analysing various details of his work, sleep and health patterns, it was found that he was likely to be only slightly fatigued.

2.4 VDR INFORMATION

The VDR information from *Samskip Courier* was analysed in detail, and provided corroborating evidence to information gained from eye witness accounts. The VDR port/starboard helm sensors were found to be connected in reverse order. This caused some initial confusion when analysing the helm movements.

2.5 VTS HUMBER

VTS Humber operatives monitored *Skagern* and *Samskip Courier* as they transited the river. It would have been inappropriate for VTS to intervene because both ships had pilots on board and they were communicating directly with each other.

The HES PMSC compliance audit in 2005 recognised the increasing demands on VTS staff resulting from growing shipping movements due to continued expansion of berth facilities on the Humber, and recommended that the watch personnel be increased

from 2 to 3 operatives. HES is fulfilling this recommendation by employing a further six operatives who should be in place by January 2007. Humber VTS equipment is also due to be renewed and upgraded at the same time. This move is welcomed, and can only serve to improve the safety of river transit.

2.6 PAVIS

The HES PAVIS held information regarding the master of *Samskip Courier*'s PEC status. He was erroneously recorded as a serving tripping master several weeks after leaving the vessel, thus having trips logged against his record. Every time a "tripping" master goes in and out of the Humber, he is required to advise the VTS by giving his "tripping number", which is only then logged on PAVIS; in this instance the system failed.

Samskip Courier was equipped with sophisticated radar, with mapping facilities, onto which a map of the Humber could be synthetically superimposed. Ships' equipment facilities can be logged on PAVIS; having such information recorded would allow pilots to be aware of, and to use, those facilities to best effect.

2.7 HUMBER PILOTAGE

The Humber estuary is a compulsory pilotage area. All vessels 60m and over are required to carry either a licensed pilot or an authorised PEC holder.

The pilots provided by HES to *Samskip Courier* and *Skagern* were fully trained, qualified and experienced.

2.8 PILOT/MASTER RELATIONSHIP

The pilot is onboard the ship in an advisory capacity; he is employed for his local knowledge of tides, currents, and topography of the land and sea bed. His role is to provide advice to the bridge team which will ensure the safe navigation of the ship. The master has overall responsibility for the safe navigation of the ship and welfare of the crew. He has the power to overrule the pilot at any time if he feels the pilot's actions may be compromising the safety of his ship. By ensuring safe navigation of the ship, the port facilities and environment are also protected.

Although the masters of *Samskip Courier* and *Skagern* were aware of their power to overrule the pilot, it appeared to be a difficult step for them to take. They put their trust in the pilots because of their expertise; they believed the pilots knew what they were doing and that there was always a valid reason for their action, although not always readily apparent to the masters. Nevertheless, the ships were the masters' responsibility, and querying pilots' actions is an appropriate and necessary part of bridge team interaction. If nothing else, the ship masters should have queried the speed of their vessels given the visibility, confined waterway and converging vessels.

Samskip Courier's master considered UK seamen to be among the finest in the world, and because of this belief, placed tremendous reliance on his pilot's ability. The master did not question the pilot's actions as the situation developed, he believed implicitly that the pilot had everything under control and, only latterly, when the pilot instructed him to go hard to port, did he query the instruction. However, he still complied.

Marine communications involving ship-pertinent information should be made on open radio frequencies so that all bridge team personnel are aware of anything which may affect them and thus play a major part in their decision making. This openness also enables all shipping in the vicinity to consider the information being exchanged and amend their own plans as necessary.

2.14 COLREGS COMPLIANCE

The COLREGs lay down rules to enhance safe navigation. These regulations apply to all vessels at all times. The Rules are quite prescriptive about actions to prevent a collision, and a regular theme throughout is the need for early avoiding action.

The vessels did not comply with Rules 2, 6, 7, 8, 19 and 35 of the COLREGs. Additionally, *Samskip Courier* did not comply with Rule 9 of these regulations.

Rule 2 requires that mariners, in the course of their duties, take *precaution(s) which may be required by the ordinary practice of seamen, or the special circumstances of the case*. The vessels met just clear of Sand End light float, which was a recognised “pinch point” in the channel, created by the positioning of the light float and sea bed topography. Although the light float did not obscure the ships from each other (had the visibility been clear), it did create a substantial obstruction which had to be negotiated. Prudent navigation would have established that the ships were going to meet in this area and appropriate action taken to avoid meeting there.

Keeping a Safe Speed: in calculating a “safe speed”, navigators must take into consideration such factors as: visibility; traffic density; vessel manoeuvrability (stopping and turning distances); background lights and the proximity of navigational hazards. Both vessels were proceeding at speeds in excess of 11kts with a closing speed of over 23kts; this could not be considered a safe speed in the prevailing restricted visibility in a narrow shallow channel with vessels converging.

Navigators are required to make proper use of radar equipment, if it is fitted and operational. Both ships had an ARPA radar. Because of frequent course alterations in the winding channel, it was believed ARPA would take too long to establish and convey information accurately, therefore it was not utilised. However, *Samskip Courier*’s radars were showing relative trails, giving a good indication of *Skagern*’s relative track (**Figure 17**). The EBL was appropriately used by the master of *Samskip Courier* which, as well as showing the vessel was clear to come onto her new heading, also gave indication of impending collision as *Skagern*’s radar target was steady on this line. The relative speed of each other’s ships could have been calculated by simple time and distance to the intervening buoy. Information was available and supplied, but not assimilated.

Rule 8 of the Regulations states, as an action to avoid collision or gain more time to assess the situation, a vessel should “*slacken her speed or take all way off by stopping or reversing her means of propulsion*”. When it became apparent that a serious situation had developed, the ships went from half ahead to slow ahead, thus taking very little way off before impact. *Skagern*’s master did put the engine controls astern, but not until *Samskip Courier* became visible at very close range, which had little effect in taking way off the vessel before colliding. Neither vessel took timely, effective avoiding action.

Rule 9, which is specifically for vessels navigating in narrow channels, requires vessels to *“keep as near to the outer limit of the channel or fairway which lies on her starboard side as is safe and practicable.”* *Samskip Courier* did not keep to her side of the channel. The vessel was allowed to cross over to apparently avoid being set onto the Sand End light float with the flood tide.

Rule 19 requires that in restricted visibility, *“every vessel shall proceed at a safe speed adapted to the prevailing circumstances and conditions.”* The Rule also gives clear instructions for vessels not in sight of one another, navigating by radar alone, to make an alteration to starboard for vessels detected forward of the beam. This did not happen.

Neither vessel was giving sound signals. Rule 35 states that, *“In or near an area of restricted visibility ... a power-driven vessel making way through the water shall sound at intervals of not more than 2 minutes one prolonged blast.”* The sound signal would have served as an indicator that the ships were closing, and possibly highlighted to the bridge teams their wayward speeds and the need for further caution. Fortuitously, the mooring party on *Skagern* was delayed from going forward – had they been forward as the two vessels approached, a sound signal from *Samskip Courier* could have given them warning to move away from the danger area.

2.15 SPEED

Both ships were proceeding at unsafe speeds for the prevailing conditions and circumstances. Had the vessels been going at slower speeds, there would have been more time for the bridge teams to react to the developing situation, therefore allowing more effective evasive action to be taken. In the event of the vessels colliding, much less damage would have occurred.

Samskip Courier's pilot explained their increase in speed (from 9kts to <12kts) after the vessel passed Saltend in terms of implied psychological pressure from the master. He felt the master's request to increase speed indicated that he wanted to make better progress. Given the lack of rapport between the pilot and master, there was no discussion about this other than the pilot's advice not to increase the vessel's speed until after they had passed Saltend jetties.

The master of *Samskip Courier* did not bring the speed down fully to slow ahead when requested to do so by his pilot because he felt that, from his experience of the vessel, a drastic speed reduction using the CP propeller might have caused her to sheer from her course. Although this might have been the case if water flow to the rudder was disturbed by the propeller blades being feathered when brought to stop, it is doubtful this would have occurred when slowing from half ahead to slow ahead. As a consequence, the pilot was not aware that the ship was not losing speed as quickly as he would have liked.

Both pilots' behaviour leading up to the collision suggests elements of complacency in continuing onwards at relatively high speed while not appreciating the approaching danger. There was little evidence of concern being raised or actions taken to acquire information and reduce uncertainty. This behaviour can arise for a number of reasons, including the failure to remain situationally aware due to a poor level of perception with respect to potential risk and a failure to project ahead. This prevented both individuals from recognising the clear signals that were present to indicate that a dangerous situation was developing until it was too late to take effective action to prevent the collision.

Perception of risk is also a factor that can influence excessive speed decisions. The pilot of *Samskip Courier* was informed by VTS, as they cleared the lock, that no other traffic was in the area. This was also heard by *Skagern*'s pilot. This might have given both pilots a sense of low risk, since the traffic in the river appeared to be minimal.

A further explanation for the excessive speed of both vessels might be habitual risk-taking. If people are accustomed to acting in a certain way, then they become less aware of the errant nature of their actions and the risks involved. This is particularly so when the behaviour has never led to negative consequences before. Habitual speeds in excess of 11kts were readily accepted by the pilots and masters on previous passages, in similar visibility.

Attention to speed might also have been inadequate because the pilots were preoccupied in monitoring their radar (but not necessarily the approaching ship). Either there was a lack of situation awareness about the vessels' speeds, or there was an acceptance of the attendant risk this produced.

All four key players recognised, in hindsight, that the ships were going too fast for the prevailing circumstances and conditions.

2.16 STEERING

Both ships were being steered by the pilots using autopilot until seconds before the collision. It is not unusual for pilots to steer vessels by autopilot, doing away with the need for a dedicated helmsman.

Many pilots prefer to use autopilot steering, as it can avoid problems caused by language difficulties or, as is sometimes the case, the helmsman's poor steering ability. However, if the pilot uses autopilot exclusively, he will lose the ability to evaluate the helmsman's language skills or steering ability for himself. The time when an emergency arises which requires manual steering, is not the best time to discover any of the helmsman's inadequacies.

Samskip Courier's lookout was also the standby helmsman. This poses the question, had he been required to serve as helmsman, who would have been lookout? The STCW Code states that the duties of the lookout and helmsperson are quite separate, and the helmsperson shall not be considered the lookout while steering, except in small ships with all round visibility from the steering position.

HES had no formal guidelines to pilots on whether a vessel should be placed in hand steering or autopilot in restricted visibility. However, there was a requirement that a helmsman be to hand should circumstances dictate, as indicated in its General Notice to Pilots No.06/2003 (**Annex 9**). MGN 315 (**Annex 6**) recommends that, during restricted visibility and hazardous navigational situations, ships should be steered by a helmsman.

2.17 PASSAGE MONITORING

Because of the dense fog, most of the navigation was being carried out by radar observation. Only during brief interludes were the bridge teams able to get visual bearings. Using radar as the primary means of navigating, the pilots simply "buoy hopped" from one buoy to the next; the ships were simply steered point to point with no use made of parallel indexing, or little use of long range scanning. The master of

Samskip Courier, however, did use his EBL appropriately to see if his ship was clear to make her next course change as she approached Sand End light float (**Figure 16**), which she was, more than 2 minutes before the collision. But because of his respect for the man's position and ability, the master did not convey this information to the pilot. He also assumed that the pilot would bring the vessel around onto her new course at any moment.

2.17.1 Skagern

Skagern had a 'port to port passage plan' applied to her conventional paper charts; the river estuary passage was monitored by the OOW checking their position with GPS and radar, annotating the chart and recording the positions in the ship's deck log.

2.17.2 Samskip Courier

The passage plan for *Samskip Courier* had been prepared to navigate the vessel from Hull port to Rotterdam port, as required by the manager's SMS.

Samskip Courier's passage plan was also plotted on her Transas 3000 ECS utilising Transas style charts which were not Electronic Navigational Chart (ENC) standard. Paper charts, which were used as the back up system, were annotated after the accident by transferring information from the ECS. Although it was not a fully approved Electronic Chart Display and Information System (ECDIS), the system was not seen as an influencing factor in the collision. *Samskip Courier*'s passage was displayed on the ECS but was not continually monitored by the master. He had no concerns with the track, which apparently followed a similar track to that taken by previous pilots, until shortly before the collision.

Although not considered an influencing factor in the collision, it should be borne in mind that using ECD systems for position monitoring cannot be relied on, particularly in estuaries where vessels accelerate and decelerate or change course quickly. Time constants set into GPS receivers and ECD systems to smooth the position can produce apparent over swings on ECDs when changing course, thus indicating the ship is out of position when she may not be. Other means should be used to cross-check positions.

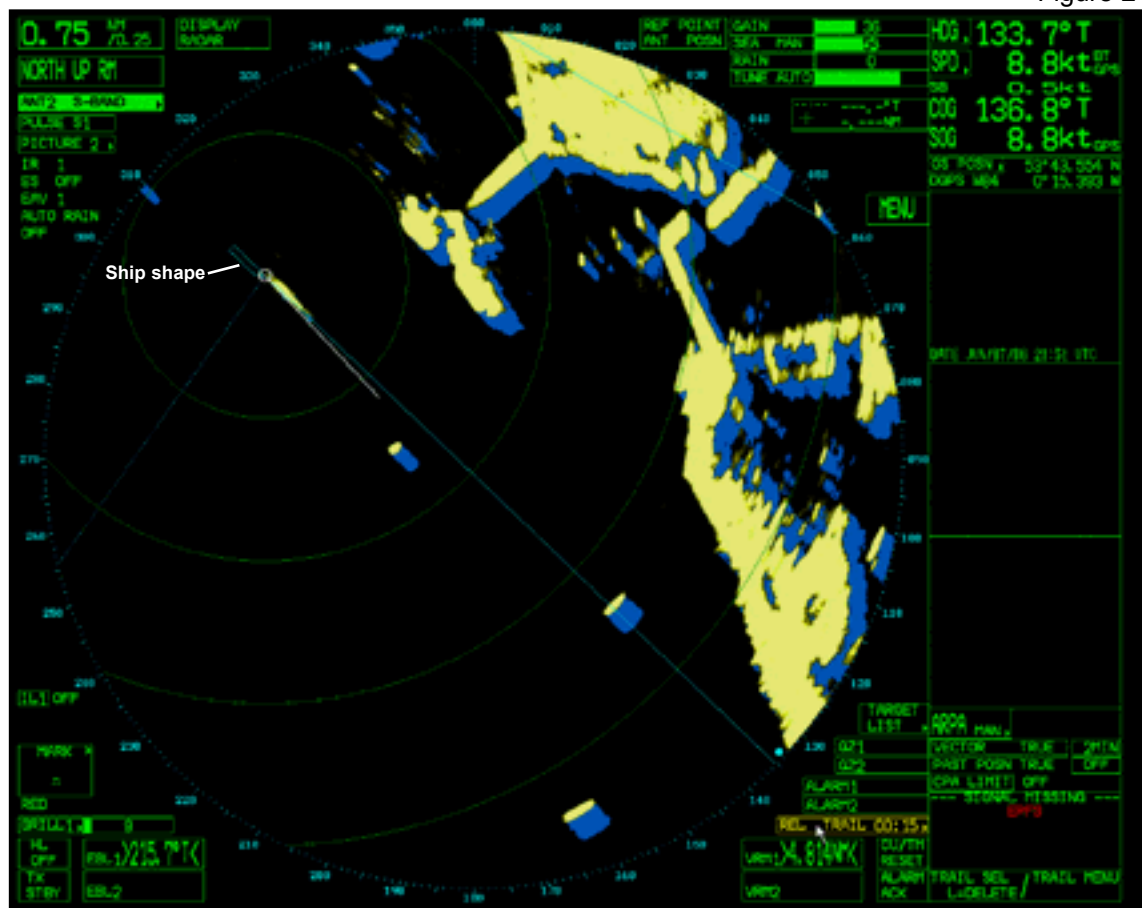
The pilot was waiting for Sand End light float to be abeam to starboard before altering course. On their general heading of 150°, *Samskip Courier* would have been approximately 1.3 cables off the light float when it was abeam. When the master applied the EBL and next course to steer, 162° at 22.56.39 (**Figure 16**) as a clearing bearing, the ship was already at an appropriate course alteration point. An alteration of 12° to starboard at this time would have brought *Samskip Courier* back to her own side of the channel and cleared the light float by approximately 0.6 of a cable.

Both radars had a mapping facility which allowed operators to synthetically superimpose coastlines, safety contours, fairways and dangers etc. Although it would not have been the pilot's duty or responsibility to create such maps, where a ship is plying the same port regularly (as *Samskip Courier* was), it would not be unreasonable to have these applied and the pilot made aware of their existence. Radar mapping facilities are excellent tools, which help observers remain aware of their position relative to surface and seabed hazards. Had the radar been used with this facility, it would have shown the pilot and master both their own and *Skagern*'s position relative to the safe water contours and surface hazards.

The port (X) band radar was being used by the pilot, while the starboard (master's) radar was S band; the radars each had decals posted alongside showing the blind sectors for the other radar (**Figures 21a, b and c**). The decal on the port radar (X band) illustrated S band blind sectors (dead astern) while the decal on the starboard radar (S band) illustrated the X band blind sectors abeam to starboard. Therefore, the pilot was using a radar which, unknown to him, had a 9° blind sector abeam to starboard and might not have shown up Sand End light float at a critical moment. This may partially explain the pilot's loss of situational awareness. Nevertheless, this would only have taken place when the target was abeam to the antenna, which was placed close to the stern of the ship. Regular observations before this time would have prevented any confusion, and a simple cross reference with the master's radar display should have allowed the pilot to clearly identify the buoy as it passed abeam.

The starboard radar display had a synthetically superimposed "ship shape" symbol showing the radar trace origin as being placed on the bow of the ship when in fact it was near the stern (**Figure 24**). It is considered that this had no bearing on the accident, but that it added an unnecessary complication to the radar picture which could provide an operator with misleading information in other circumstances.

Figure 24



Synthetic ship shape symbol showing radar scanner positioned on bow of *Samskip Courier*

2.18 POSITION OF COLLISION

Shoal water areas in the Humber shift continually due to the effects of currents and tides. HES carries out regular surveys of the river to monitor the positions of these shallow areas and place buoyage accordingly. The Sand End light float was moored where the Hull Middle sands traditionally ended. Bottom surveys carried out by HES showed that there had been relatively deep water to the west of the buoy for over a year before the accident. The PMSC recommends that the harbour authority '*place and maintain navigation marks where they will be of best advantage to navigation*' and '*to keep a vigilant watch for any changes to sea or river bed affecting the channel or channels and move or renew sea navigation marks as appropriate.*^[sic]

The 7 metre contour is a fairly good indicator of safe water in this area - once less than 7 metres, the water shoals rapidly on each side of the channel. The Sand End light float was set fully 1 cable out from the 7 metre contour of the Hull Middle Sands. The positioning of the buoy effectively created a 'knuckle' and narrowing in the channel, to as little as 2 cables at its narrowest point. A straight line between Elbow and Skitter Haven buoys shows the 7 metre contour breaking this line occasionally, just as it does between Paull Sand buoy and No 19A on the other side of the channel (**Figure 4**). There is currently room for the light float to be closer to the shallows, thus widening the navigable channel and removing the knuckle.

Many pilots recognize that the Sand End light float can be slipped (passed on the wrong side) if need be. Nevertheless, it should be best placed in the channel to reduce the need for such a manoeuvre. Had the pilot or master of *Samskip Courier* appreciated the risk posed by the approaching *Skagern* earlier, it would have been quite safe to place the vessel to the starboard side of the river and "slip" the Sand End light float in order to pass well clear of *Skagern*.

In the immediate area where the collision took place, there were three gas pipelines running from Skitter Haven to Thorngumbald. These pipes, owned by Transco and Centrica, vary from between 24 to 42 inches in diameter. When originally laid, the pipes were buried but, over the years, they have become exposed in places due to currents and tidal influence scouring the seabed. Regular surveys are carried out on the pipelines, and when they are discovered to be under scoured or exposed they are sand bagged. This then encourages natural sand deposits to build over them. Pilots are forcefully advised by HES of the dangers of anchoring in the vicinity of the pipes or of damaging them for any reason. A notice to this effect is also highlighted on the chart.

2.19 DECISION MAKING AND CONTINGENCY PLANNING

As it became apparent that a serious situation had developed, the pilots and masters of the vessels failed to take the appropriate evasive action of taking all way off their vessel, by going astern. Both ships belatedly went to slow ahead, but carried their way until impact. There was an attempt by both pilots to communicate their way out of trouble, as indicated in *Skagern*'s pilot's request for a radio channel change, and *Samskip Courier*'s pilot's attempt to do so.

Given the already close proximity and position of the vessels when the first radio exchange was made highlighting the gravity of their situation, an immediate decision should have been taken, by both pilots, to take all way off. While one pilot fumbled to

select a new radio channel, the other made requests for him (which went unheard) to turn his vessel to starboard. Both were hoping for the other to do something to resolve the situation. Neither made the ultimate decision of taking all way off their vessel.

Another option would have been to either run the ships ashore or, in the case of *Samskip Courier*, hit the Sand End light float. Either option would have been better than colliding with each other, and would have caused considerably less damage. For such a decision to be made, some thought would have to be given beforehand to various emergency scenarios and their contingency planning. It would not have been unreasonable for pilots operating these waters on a daily basis to have given thought to such contingencies as part of a personal risk assessment. If the gas pipelines in the vicinity were considered a danger to running ashore, then all the more reason to be going at a reduced speed in this area and avoid passing movements there.

The masters of both ships had great difficulty in taking the con from their pilots as they knew the two (pilots) were communicating with each other. Again, as part of contingency planning, this is something masters should be asking before it happens – ‘*at what stage should I step in and take conduct of my ship?*’ A reasonable answer to this would be, ‘*after I have made relevant observations and spoken to the pilot regarding his actions.*’ Although the masters observed the incident unfolding, neither verbally queried their ship’s speed or (lack of) action being taken. By not physically querying their pilots’ actions, the masters’ decision to take the con was all the more difficult.

2.20 TIDAL INFLUENCE

Samskip Courier left King George Dock some 21 minutes after low water at the dock. The pilot informed the master it was 40 minutes after low water and that a flood tide would be running during the passage. When the flood tide is running there is a pronounced set onto Sand End light float, normally recognised to affect inbound ships (as they slow down for Saltend jetties), more so than those proceeding outbound.

At 22:55:10 *Samskip Courier*’s pilot gave the vessel 10.5° port helm as the radar vector showed them to be setting towards the Sand End float, which the pilot felt was being caused by the tide. However, examination of the VDR showed that at 22:53:40, starboard helm of as much as 9.6° was given and held for 50 seconds as they passed Elbow buoy; the subsequent port helm appears to be a correction for the effect of the earlier starboard helm and not the effect of the tide as the pilot believed. Examination of the radar plots shows the ship’s radar vector consistently heading towards Paull Sand buoy once past Elbow buoy (**Figures 12-19**). The evidence suggests that the general heading was just to the north of Paull Sand buoy until just prior to collision. There does not appear to be evidence from the radar information of pronounced setting towards the Sand End buoy, and the images confirm that *Samskip Courier* was mid channel on the transit between Elbow and Sand End buoys.

By 22:56 *Samskip Courier* was on the north side of the channel to counteract the perceived tidal set, but it was effectively slack water, with little or no cross channel set. Three minutes after the collision, *Samskip Courier* was lying stopped in the water (the head on collision stopped both ships almost instantly). Her COG and SOG at this time were 265°, 0.3kts; as taken from VDR information after the instruments had settled. This movement was effectively the set and drift caused by the tide and wind at the